

## 2.9 Dimension and Rank

I

Dimension of a vector space: The dimension of a finite vector space  $V$  is the number of vectors  $\neq \vec{0}$  in the basis of  $V$ .

Rank of a matrix  $A$  is the dimension of  $\text{Col } A$ .  $\text{Rank } A := \dim \text{Col } A$ .

Drawing on an earlier example ("Example both null and column space"), we get:

$$A = \begin{bmatrix} 3 & 1 & 1 & 6 \\ 1 & -2 & 5 & -5 \\ 4 & 1 & 2 & 7 \end{bmatrix} \quad \text{g.f.} \quad \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -2 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

columns 1 and 2 are linearly independent and form a basis for  $\text{Col } A$ .

$$\text{Rank } A := \dim \text{Col } A = 2$$

Columns 3 and 4 are the free variable columns, and form the basis for  $\text{Nul } A$ .

$$\dim \text{Nul } A = 2 \quad (\text{Also called the nullity of } A)$$

Theorem 14 - The Rank Theorem

If a matrix  $A$  has  $n$  columns, then  $\text{Rank } A + \dim \text{Nul } A = n$

## 2.9 Dimension and Rank (cont'd)

II

### Example 1

Let  $\vec{v}_1 = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$ ,  $\vec{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ ,  $\vec{x} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$ ,  $B = \{\vec{v}_1, \vec{v}_2\}$ . Then  $B$  is a basis for  $H = \text{span}\{\vec{v}_1, \vec{v}_2\}$  because

$\vec{v}_1$  and  $\vec{v}_2$  are linearly independent. Determine if  $\vec{x}$  is in  $H$ , and if it is, find the coordinate vector of  $\vec{x}$  relative to  $B$ .

### Solution:

If  $\vec{x}$  is in  $H$ , then we must be able to say  $c_1\vec{v}_1 + c_2\vec{v}_2 = \vec{x}$ . Meaning there must exist a  $c_1$  and  $c_2$  so that the equation holds true. We can find this through row operations.

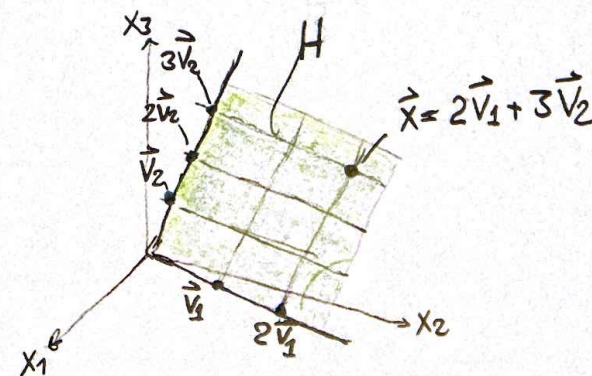
$$\begin{bmatrix} 3 & -1 & 3 \\ 6 & 0 & 12 \\ 2 & 1 & 7 \end{bmatrix} \xrightarrow{\text{Matlab}} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{Row reduction shows that } 2\vec{v}_1 + 3\vec{v}_2 = \vec{x}. \text{ Let's test it.}$$

$$2\begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix} + 3\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$$

$$c_1 = 2, c_2 = 3, [\vec{x}]_B = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & -1 \\ 6 & 0 \\ 2 & 1 \end{bmatrix} \cdot [\vec{x}]_B \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix} \quad \left. \vphantom{\begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}} \right\} \text{Linear transformation}$$

The dimension of a non zero subspace  $H$ , denoted by  $\dim H$ , is the number of vectors in any basis for  $H$ . The dimension of the zero subspace  $\{\vec{0}\}$  is defined to be zero.



The basis  $B$  determines a "coordinate system" on  $H$ , visualized in the figure.

## 2.9 Dimension and Rank (cont'd)

III

Definition The rank of a matrix  $A$ , denoted by  $\text{rank } A$ , is the dimension of the column space of  $A$ .

Example

$$A = \begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 4 & 7 & -4 & -3 & 9 \\ 6 & 9 & -5 & 2 & 4 \\ 0 & -9 & 6 & 5 & -6 \end{bmatrix} \sim \begin{bmatrix} \overset{P}{2} & \overset{P}{5} & -3 & -4 & 8 \\ 0 & -3 & 2 & 5 & -7 \\ 0 & 0 & 0 & 4 & -6 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$p$  = pivot columns;  $A$  has 3 pivot columns, thus its  $\text{rank} = 3$ , because the pivot column vectors form a basis for  $\text{Col } A$ .

Theorem 14 - The Rank Theorem

If a matrix  $A$  has  $n$  columns, then  $\text{rank } A + \dim \text{Nul } A = n$ .

Theorem 15 - The Basis Theorem

Let  $H$  be a  $p$ -dimensional subspace of  $\mathbb{R}^n$ . Any linearly independent set of exactly  $p$  elements in  $H$  is automatically a basis for  $H$ . Also, any set of  $p$  elements of  $H$  that spans  $H$  is automatically a basis for  $H$ .

## 2.9 Dimension and Rank (cont'd)

IV

### The Invertible Matrix Theorem

Let  $A$  be a square  $n \times n$  matrix. Then the following statements are equivalent. That is, for a given  $A$ , they are either all false or all true.

- a.  $A$  is an invertible matrix.
- b.  $A$  is row equivalent to the  $n \times n$  identity matrix.
- c.  $A$  has  $n$  pivot positions.
- d. The equation  $A\vec{x} = \vec{0}$  has only the trivial solution.
- e. The columns of  $A$  form a linearly independent set.
- f. The linear transformation  $\vec{x} \mapsto A\vec{x}$  is one-to-one.
- g. The equation  $A\vec{x} = \vec{b}$  has at least one solution for each  $\vec{b}$  in  $\mathbb{R}^n$ .
- h. The columns of  $A$  span  $\mathbb{R}^n$ .

- i. The linear transformation  $\vec{x} \mapsto A\vec{x}$  maps  $\mathbb{R}^n$  onto  $\mathbb{R}^n$ .
- j. There is an  $n \times n$  matrix  $C$  such that  $CA = I$ .
- k. There is an  $n \times n$  matrix  $D$  such that  $AD = I$ .
- l.  $A^T$  is an invertible matrix.
- m. The columns of  $A$  form a basis of  $\mathbb{R}^n$ .
- n.  $\text{Col } A = \mathbb{R}^n$ .
- o.  $\text{rank } A = n$ .
- p.  $\dim \text{Nul } A = 0$ .
- q.  $\text{Nul } A = \{\vec{0}\}$ .